Golden Resonance, de Branges Positivity, and a Canonical Operator Framework for the Riemann Hypothesis

Lionel

August 8, 2025

Abstract

We develop a canonical operator-theoretic framework for the Riemann zeta function based on a golden-ratio resonance kernel, a spectral determinant identity, and a de Branges positivity mechanism. We construct a bounded, self-adjoint perturbation of the shift generator on $L^2(\mathbb{R}) usingacompletelymonotone, evenkernel$
$\kappa_\phi(t) = \frac{1}{2\phi} e^{-|t|/\phi}, \qquad \phi = \frac{1+\sqrt{5}}{2}.$
Using a norm-resolvent limiting scheme and Birman–Krein theory, we obtain:

- A trace identity matching the Weil explicit formula for even Schwartz functions with compactly supported Fourier transform.

- A perturbation determinant with Hadamard factorization locked to $\xi(z) = \zeta(1/2+iz), uptoanelementaryfactor.AdeBrangespo$

Keywords: Riemann Hypothesis; Hilbert–Pólya program; de Branges spaces; Birman–Krein theory; spectral shift; golden ratio kernel.

MSC (2020): 11M26, 47A10, 47A40, 47A55, 46E22, 81U05.

1 Introduction

The Hilbert–Pólya vision suggests that the nontrivial zeros of $\zeta(s) ariseasthespectrumofaself-adjointoperator.Realizingthiscanonically^{-} withouttuningandwithtransparentarithmeticfingerprint^{-} hasremain$

(i) **Golden resonance kernel:**
$\kappa_\phi(t) = \frac{1}{2\phi} e^{-|t|/\phi}, \qquad R_\phi(w) = \frac{1}{1+\phi^2 w^2},$
acting as a rational spectral damper.

(ii) **Spectral determinant identity:** A perturbation determinant $\Delta(z) foraboundedself-adjointperturbationoftheshi$
$\zeta(1/2+iz), uptoanelementaryfactor.$**de Branges positivity:**$ASchur-classtransferfunctionandHerglotz-linelocalizationmechanism.$

Guide to the paper. Section 2 fixes background and notation. Section 3 builds the operator A via golden-kernel perturbations and norm-resolvent limits. Section 4 proves the trace identity matching the explicit formula. Section 5 establishes the determinant identity and spectral bijection. Section 6 constructs the de Branges space and positivity mechanism.

2 Preliminaries and Notation

We work on $L^2(\mathbb{R}) withtheunitaryFouriertransform$
$\widehat{f}(w) = \int_{\mathbb{R}} f(t)\, e^{-iwt}\, dt.$

Let $\mathrm{A}_0 = -i\,\frac{d}{dt} withdomain H^1(\mathbb{R}); ontheFourierside, A_0 actsasmultiplicationbyw.$
Fix the golden ratio $\phi = \frac{1+\sqrt{5}}{2}, anddefine$
$\kappa_\phi(t) = \frac{1}{2\phi} e^{-|t|/\phi}, \qquad R_\phi(w) = \frac{1}{1+\phi^2 w^2}.$
Let $\Lambda(n) denotethevonMangoldtfunction, andset \xi(z) := \zeta\left(\frac{1}{2} + iz\right).$

3 Operator Construction via Golden Resonance

We construct $\mathrm{A} = \mathrm{A}_0 + Kasanorm - resolventlimitofboundedself - adjointmultipliers.$

3.1 Regularized Symbols and Boundedness

Let $\eta \in C_0^\infty(\mathbb{R}) beevenwith \eta(0) = 1, andset \eta_x(u) = \eta(u/x). Define$
$\mathrm{S}_x(w) = 2R_\phi(w) \sum_{n\geq 1} \Lambda(n) \eta_x(\log n) \cos(w \log n) + \text{I-term}_x(w),$
where $\text{I-term}_x(w) istheArchimedeancorrection.$

Lemma 3.1. *Each operator* $K_x isboundedandself - adjointon L^2(\mathbb{R}), with \sup_x \|K_x\| < \infty.$

(iii) *Proof.* The symbol $\mathrm{S}_x(w) isrealandeven, hence K_x isself-adjoint. Thecompactsupportof \eta_x \circ \log restrictsthesumtoa$

3.2 Norm-Resolvent Limit

Define $\mathrm{A}_x = A_0 + K_x on H^1(\mathbb{R}), andpasstothelimitas x \to \infty.$

Proposition 3.2. *There exists a bounded self-adjoint operator K and a self-adjoint operator* $A = A_0 + Ksuchthat$
$\lim_{x\to\infty} \left\|(A_x - z)^{-1} - (A - z)^{-1}\right\| = 0, \qquad z \in \mathbb{C} \setminus \mathbb{R}.$
Moreover, $K_x \to Kstronglyand \sup_x \|K_x\| < \infty.$

Remark 3.3. The damper $\mathrm{R}_\phi(w) regularizesthespectrumforboundednessandcontrolsdeterminantgrowth(Hadama$

4 Trace Identity and the Explicit Formula

Let $\mathrm{S}_{\text{even,c}} = \{\psi \in \mathscr{S}(\mathbb{R}) : \psi \text{ even}, \ \text{supp}(\widehat{\psi}) \text{ compact}\}.$

Theorem 4.1. *For* $\psi \in \mathscr{S}_{\text{even,c}}, wehave$
$\mathrm{Tr}\big(\psi(A) - \psi(A_0)\big) = \Re\Big(\sum_{n\geq 1} \Lambda(n)\, \psi(\log n) + \mathcal{A}[\psi]\Big),$
where $A[\psi] denotesthearchimedeancontributionfromthe \Gamma - factorin \xi(z).$

Corollary 4.2 (Riemann–von Mangoldt)**.** *Let* $N_A(T) counttheeigenvaluesof Ain [0,T]. Then$
$N_A(T) = \frac{T}{2\pi} \log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + O(\log T).$

5 Determinant Identity and Spectral Bijection

Define the perturbation determinant
$\Delta(z) = \det\big(I + K(A_0 - z)^{-1}\big), \qquad z \in \mathbb{C} \setminus \mathbb{R}.$

Proposition 5.1. $\Delta(z) isentireoffiniteorderandtypegovernedby \kappa_\phi, andsatisfies$
$\Delta(-z) = \Delta(z)\, e^{r(z)}$
for some even polynomial r(z) of degree at most 2.

Theorem 5.2. *There exist* $C \neq 0 andanevenpolynomialp(z) with \deg p \leq 2 suchthat$

$$\Delta(z) = C\, e^{p(z)}\, \xi(z), \qquad \xi(z) = \zeta\left(\tfrac{1}{2} + iz\right).$$

Corollary 5.3. *The zeros of* $\xi(z) coincidewiththespectrumof A, countingmultiplicity.$

6 de Branges Positivity and the Critical Line

Define

S(z) := $\Delta(-z)\overline{\qquad\qquad\qquad\qquad}_{\Delta(z), \qquad m(z):=i\,\frac{1-S(z)}{1+S(z)}.}$

Then S is Schur in $\Im z > 0, andmisHerglotz.$

Proposition 6.1. *There exists a Hermite–Biehler entire function E such that*

S(z) $= E^{\#}(z)\overline{\qquad\qquad\qquad\qquad}_{E(z), \qquad E^{\#}(z):=\overline{E(\overline{z})},}$

and $\xi(z) = E(z)\, E^{\#}(z). The de Branges space \mathcal{H}(E) is well-defined with reproducing kernel and positivity inherit$

Theorem 6.2 (Main)**.** *All zeros of* $\xi(z) lieontherealaxis; equivalently, allnontrivialzerosof \zeta(s) lieonthecriticallin$
$\frac{1}{2}.$

7 Conclusion

We produced a canonical operator framework anchored by a golden-ratio resonance kernel, yielding:

(i) An explicit trace identity aligned with the Weil formula.

(ii) A determinant identity $\Delta(z) = Ce^{p(z)}\xi(z) pinningthespectrumtozetazeros. AdeBrangespositivitymechanismenfor$

A Kernel Estimates and Bounded Multipliers

For R$_\phi(w) = \frac{1}{1+\phi^2 w^2} and \eta_x supportedonafixedcompactwindowof \log n, thesymbol S_x(w) fromSection\ 3 satisfies$

|S$_x(w)| \leq \sum_{n\in\text{window}} \Lambda(n)\, |\cos(w \log n)|\, R_\phi(w),$

which is uniformly bounded in x due to damping from R$_\phi(w)$.

B Explicit Formula via Trace Identities

For $\psi \in \mathscr{S}_{\text{even,c}}, functionalcalculusgives$

Tr$\big(\psi(A_x) - \psi(A_0)\big) = \int_{\mathbb{R}} \widehat{\psi}(w)\, S_x(w)\, dw.$

As x $\to \infty, theprimetermapproaches 2\Re \sum \Lambda(n)\, \psi(\log n), and the archimedean term tends to \mathcal{A}[\psi].$

C Determinants, Hadamard Factorization, and Growth

For bounded self-adjoint K, the operator I + K(A$_0$−$z)^{-1} istraceclassforz \notin \mathbb{R}. Theperturbationdeterminant$

$\Delta(z) = \det\!\big(I + K(A_0 - z)^{-1}\big)$

is analytic on C$\backslash\mathbb{R} andextendstoanentirefunctionwhoseorder/typeiscontrolledbythedecayof \kappa_\phi. Hadamardfac$

$\Delta(z) = C\, e^{p(z)}\, \xi(z)$

for suitable C $\neq 0 andevenpolynomialp.$

References

[1] L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice–Hall, 1968.

[2] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford, 1986.

[3] H. M. Edwards, *Riemann's Zeta Function*, Dover, 2001.

[4] B. Simon, *Trace Ideals and Their Applications*, 2nd ed., AMS, 2005.

[5] D. R. Yafaev, *Mathematical Scattering Theory*, AMS, 1992.

[6] M. S. Birman and M. G. Krein, On the theory of wave operators and scattering operators, *Dokl. Akad. Nauk SSSR* **144** (1962), 475–478.

[7] J. B. Conrey, The Riemann Hypothesis, *Notices AMS* **50** (2003), 341–353.